

Reasoning Reference: Cubes and Dice

Educational Series

December 29, 2025

1 Introduction

Cubes and Dice problems primarily focus on identifying the spatial relationship between faces. A standard die has 6 faces, where the sum of opposite faces is usually 7, though reasoning problems often use "General Dice" with arbitrary markings.

2 Rules for Closed Dice

When comparing two positions of the same die, use these rules to find opposite faces:

2.1 Rule 1: One Common Face

If two positions show one common face in the same position, then the other corresponding faces are opposite to each other. *Alternative (Clockwise Rule)*: If the common face is in different positions, write the faces in clockwise order starting from the common face. The pairs aligned will be opposites.

2.2 Rule 2: Two Common Faces

If two positions show two common faces, the remaining third faces in both positions are opposite to each other.

- **Example:** Position 1: (2, 3, 4); Position 2: (2, 3, 5). Here, 4 is opposite to 5.

3 Open Dice

In an open die (a net), the faces that are **alternate** to each other (separated by one square) are opposite.

- Faces in a straight line: The 1st and 3rd faces are opposite; the 2nd and 4th faces are opposite.
- The remaining two side-flaps are opposite to each other.

4 Painted Cube Problems

When a large cube of side L is painted and cut into smaller cubes of side s , let $n = L/s$ (number of divisions). The distribution of smaller cubes is:

1. **3 Faces Painted (Corner Cubes):** Always **8** (regardless of n).
2. **2 Faces Painted (Edge Cubes):** $12(n - 2)$
3. **1 Face Painted (Central Cubes):** $6(n - 2)^2$
4. **0 Faces Painted (Inner/Core Cubes):** $(n - 2)^3$
5. **Total Small Cubes:** n^3

5 Solved Examples

Example 1: Dice Rotation Two positions of a die are shown: (1, 2, 3) and (1, 4, 5). Find the face opposite to 2.

- **Step 1:** Common face is 1.
- **Step 2:** Clockwise from 1 in Pos 1: $1 \rightarrow 2 \rightarrow 3$.
- **Step 3:** Clockwise from 1 in Pos 2: $1 \rightarrow 4 \rightarrow 5$.
- **Conclusion:** 2 is opposite to 4, and 3 is opposite to 5.

Example 2: Painted Cube A cube painted red on all faces is cut into 64 small identical cubes. How many have exactly one face painted?

- $n^3 = 64 \implies n = 4$.
- Formula for 1 face painted: $6(n - 2)^2$.
- Calculation: $6(4 - 2)^2 = 6(2^2) = 6 \times 4 = 24$.
- **Answer:** 24 cubes.